

题目. *Hausdorff dimension of recurrence sets*

摘要. We consider linear mappings on the d -dimensional torus, defined by $T(x) = Ax \pmod{1}$, where A is an invertible $d \times d$ integer matrix, with no eigenvalues on the unit circle. In the case $d = 2$ and $\det A = \pm 1$, we give a formula for the Hausdorff dimension of the set

$$\{x \in \mathbb{T}^d : x \in B(T^n(x), e^{-\alpha n}) \text{ for infinitely many } n\}.$$